Moussa Traoré

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Portfolio: musananko.github.io | GitHub: github.com/MusaNanko

Software Engineer: Backend, Data Platforms and Automation

Software engineer with 2 years of experience building data platforms, back-end services and web apps for research in Python, Java and TypeScript. I build sensor-data pipelines and monitoring platforms, run single sign-on and CI/CD for multi-service systems, and presented my work at JDEV 2026, a national software conference.

Skills

- **Languages:** Python | Java | C++ | SQL | TypeScript / JavaScript | R | Bash
- **Backend and Data:** FastAPI | Flask | Java EE | REST APIs | PostgreSQL | TimescaleDB | MySQL | Oracle | Airflow
- **Frontend:** React | Angular | R Shiny | Vite | Tailwind CSS
- **DevOps and Quality:** Docker | GitLab CI/CD | Git | Linux | Traefik | pytest | Trivy
- **Identity and Security:** Keycloak | OpenID Connect | single sign-on | vulnerability management
- **Languages:** French (native) | Bambara (native) | English (professional working proficiency)

Work Experience

French National Centre for Scientific Research (CNRS) — Besançon, France
Software Engineer, Data Platform — *02/2025 – Present*

- **Sens2viz:** monitored platform that retrieves, stores, visualizes and shares sensor data under FAIR principles, built as Dockerized microservices on Airflow, TimescaleDB and Grafana. Presented at JDEV 2026, a national conference.
- **Data pipelines:** Python Airflow pipelines ingesting 10M+ sensor records, with validity checks and pytest tests.
- **Identity:** Keycloak (OpenID Connect) single sign-on for 10 services behind Traefik, with separate access per sensor network.
- **KarstFlow:** built a FastAPI and React app that detects and corrects karst sensor artefacts without altering real signal.
- **OMIV:** maintain the R Shiny apps, Flask back-ends and configuration of the French Landslide Observatory.
- **Performance and releases:** up to 60% faster SQL queries; tag-based GitLab CI/CD releases with blocking Trivy scans.

AI Mali — Remote
Early Member (volunteer) — *2025 – Present*

- **Front end:** built the front end (React, Vite, Tailwind) of the bilingual website and CMS, one of three contributors.
- **Weekly Talks:** chaired about 30 of the 40 online talks with researchers and policymakers; speaker of Talk #30.
- **TimaTon:** technical choices for an offline-first reforestation app; setting up its iOS release.

Atlantic Wetlands Forum — Rochefort, France
Software Engineer — *08/2024 – 01/2025*

- **Software quality:** introduced unit tests and verified deployment: 25% fewer production bugs, 40% faster deployments.
- **Backend:** Java EE web services and Angular front end for a national environmental data platform.

Atlantic Wetlands Forum — Rochefort, France
Software Engineering Intern — *02/2024 – 08/2024*

- **Specifications:** 15 requirements for a national Oracle database and the architecture for 50,000+ datasets.

Talks and Publications

- **Sens2viz: a monitored platform for retrieving, managing and visualizing sensor data** | JDEV 2026 | co-author (link)
- **Breaking the Barriers of Text-Hungry and Audio-Deficient AI** | arXiv, 2025 | co-author (link)

Education

MSc in Computer Science, Databases and Artificial Intelligence — 2022 – 2024
University of Burgundy (Université de Bourgogne), Dijon, France

BSc in Mathematics and Computer Science — 2019 – 2022
University of Burgundy (Université de Bourgogne), Dijon, France

Projects

- **Real-time ray marching engine** | C++, OpenGL, GLSL | github.com/MusaNanko/MGSI-Ray-Marching: interactive GPU renderer for scenes built from signed distance functions in a CSG tree, with depth of field and motion blur.